

Lab Assignment

Subject: Problem Solving using Data Structures Lab [TIU-ES-UCS-L12101]

1st Year, 2nd Semester

Assignment – 4

1. Create a Singly non-circular linked list and perform insertion (beginning, middle, end) of a new node and deletion (beginning, middle, end) of a node. Display the linked list after each operation.
2. Create a Singly circular linked list and perform insertion (beginning, middle, end) of a new node and deletion (beginning, middle, end) of a node. Display the linked list after each operation.
3. Create a Doubly non-circular linked list and perform insertion (beginning, middle, end) of a new node and deletion (beginning, middle, end) of a node. Display the linked list after each operation.
4. Create a Doubly non-circular linked list and perform insertion (beginning, middle, end) of a new node and deletion (beginning, middle, end) of a node. Display the linked list after each operation.
5. Write a program to find any loop present in the linked list.
6. Write a program to find the node which is in the middle position of a linked list.
7. Write a program to Reverse the order of elements in a singly linked list.
8. Write a program to Merge two linked lists into one list.
9. Write a program to Implement a sparse matrix using linked list.
Represent a polynomial expression using linked list and perform addition on 2 such polynomial expressions.